CMPG 325 Computer Networks: Technical Report

Dikgosi History & Archives Research Centre (DHARC), Mafikeng: Final Submission (Milestone 3)

Item	Detail
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Project ID / Client ID	CMPG325-2026-112 / CLI-112
Assigned addressing block	10.42.0.0/16
Assigned feature	Internal HTTP web server in VLAN 40, reachable only from VLAN 10 and VLAN 20
Change request	CR9: secure remote management by one off-site administrator (SSH, source-restricted)
Software	Cisco Packet Tracer 9.0.1
Submission date	16 October 2026
GitHub portfolio	https://github.com/TebelloQekisi/cmpg325-2026-112-dharc-network.git

1. Executive summary

DHARC catalogues, digitises and provides access to historical records. It needs a network that serves administrative staff, archivists and researchers, and a public Reading Room, while hosting an internal web portal that the public must never reach. The brief also requires secure remote management for one off-site administrator (CR9) and no downtime during month-end processing.

This report documents the final solution: a collapsed-core campus network built in Cisco Packet Tracer, using one edge router, one multilayer core switch and one Reading Room access switch. Five VLANs separate Administration, Archives/Research, the Reading Room, Servers and Management. A two-link EtherChannel protects the core-to-access path. The assigned feature, an internal web server restricted to VLANs 10 and 20, is enforced by an extended ACL on the core. All 21 planned tests produced the expected behaviour.

The report follows the project from the client problem through design, implementation, testing and reflection, and it records where the final build differs from the Milestone 1 design and why.

2. Client requirements

2.1 Stakeholders

Group	Description	Network need
Administration	Finance, HR, front-office staff	General LAN access
Archives & Research	Historians, archivists, digitisation staff	LAN access and the internal web portal
Reading Room (public)	Walk-in researchers on public terminals	Isolated; no access to staff or server VLANs
Off-site administrator (CR9)	Remote network administrator	Secure SSH management of network devices only

The brief does not give headcounts, so two representative hosts per department are modelled. Each VLAN uses a /24, which leaves room to grow without re-addressing.

2.2 Functional requirements

1. Full IP connectivity between authorised departments using 10.42.0.0/16.
2. An internal HTTP server reachable by Administration and Archives/Research only.
3. VLAN segmentation so Reading Room traffic cannot reach staff or server subnets.
4. SSH-only remote management for the off-site administrator, restricted by source address, with Telnet unusable.
5. DHCP for end-user VLANs; static addresses for infrastructure and the server.
6. A working, testable Packet Tracer implementation.

2.3 Non-functional requirements

- **Availability:** no downtime during month-end processing (brief section 8), addressed by link redundancy between the core and access switch.
- **Security:** encrypted, source-restricted management; public and staff traffic in separate broadcast domains.
- **Scalability:** the /16 block leaves substantial room for future departments.
- **Maintainability:** consistent VLAN-to-department mapping and a documented addressing scheme.

2.4 Assumptions and scope

- Full WAN or ISP-side redundancy (dual links, HSRP) is out of scope; the downtime constraint is addressed at the LAN layer, where cable and port failures are the most likely cause of local outages.
- The web server is internal only and is not exposed to the internet.
- Headcounts, floor plan and cable lengths are assumed for this simulation.

3. Network design

3.1 Physical topology

A collapsed-core design was chosen. The edge router handles only the link to the off-site administrator, the multilayer core switch performs all inter-VLAN routing, and a separate access switch keeps the public Reading Room apart from staff and server traffic.

Device	Model	Role
RTR-EDGE	Cisco 4321 (ISR4321)	Edge router; link to the off-site administrator and to the core
CORE-L3SW	Cisco 3650-24PS multilayer switch	Inter-VLAN routing, DHCP server, ACL enforcement, staff and server access
ACC-READSW	Cisco 2960-24TT	Reading Room access switch
WEB-SRV	Server-PT	Internal HTTP web server (VLAN 40)
PC-Admin1, PC-Admin2	PC-PT	Administration workstations (VLAN 10)
PC-Archivist1, PC-Archivist2	PC-PT	Archives and research workstations (VLAN 20)
PC-Reading1, PC-Reading2	PC-PT	Public Reading Room terminals (VLAN 30)
Remote-Admin-PC	PC-PT	Off-site administrator (192.0.2.2/30)

Cabling as built

From	To
Remote-Admin-PC Fa0	RTR-EDGE Gi0/0/0
RTR-EDGE Gi0/0/1	CORE-L3SW Gi1/0/6
CORE-L3SW Gi1/0/23 and Gi1/0/24	ACC-READSW Fa0/23 and Fa0/24 (Port-channel 1)
CORE-L3SW Gi1/0/1	WEB-SRV Fa0
CORE-L3SW Gi1/0/2, Gi1/0/3	PC-Admin1, PC-Admin2
CORE-L3SW Gi1/0/5, Gi1/0/4	PC-Archivist1, PC-Archivist2
ACC-READSW Fa0/3, Fa0/4	PC-Reading1, PC-Reading2

Key physical design decisions

- **Multilayer core switch for inter-VLAN routing (SVIs):** local VLAN-to-VLAN traffic never reaches the WAN-facing router, which keeps the router simple and the routing fast.
- **Dedicated Reading Room access switch:** public traffic never shares an access device with staff or archive traffic.
- **Two-link EtherChannel (Po1):** removes the single cable as a point of failure between the core and the Reading Room, addressing the month-end availability constraint without the cost and complexity of a second router.

3.2 Logical topology and IP addressing

VLAN	Name	Subnet	Gateway (core SVI)	Addressing
10	Administration	10.42.10.0/24	10.42.10.1	DHCP (.1 to .9 excluded)
20	Archives-Research	10.42.20.0/24	10.42.20.1	DHCP (.1 to .9 excluded)
30	Reading-Room	10.42.30.0/24	10.42.30.1	DHCP (.1 to .9 excluded)
40	Servers	10.42.40.0/24	10.42.40.1	Static: WEB-SRV 10.42.40.10
99	Management	10.42.99.0/24	10.42.99.1	Static: ACC-READSW 10.42.99.2

Link	Subnet	Addresses
RTR-EDGE to CORE-L3SW (transit)	10.42.0.0/30	Router Gi0/0/1 10.42.0.1; core Gi1/0/6 10.42.0.2
RTR-EDGE to off-site administrator	192.0.2.0/30	Router Gi0/0/0 192.0.2.1; Remote-Admin-PC 192.0.2.2

The assigned /16 is divided into five /24 VLAN subnets plus a /30 transit link taken from the same block, with no overlap. The off-site link uses the documentation range 192.0.2.0/24 (RFC 5737) as a stand-in for a real public address, because 10.42.0.0/16 is private.

VLSM reasoning. Each department fits comfortably in a /24, so variable-length subnets would save no useful address space. Uniform /24s are simpler to explain, verify and test. The /30 point-to-point links show VLSM used where it is appropriate, since only two hosts are needed.

3.3 Design justification summary

Decision	Why

Decision	Why
Collapsed core	Small campus; one core and one access switch meet the requirements without unnecessary layers
SVIs on the core	Fast local inter-VLAN routing; keeps the router out of local traffic
Separate VLAN for servers	One place to apply the web-server access policy
Management VLAN 99	Keeps device management apart from user traffic
Static routing	One edge router and one core; a routing protocol adds complexity without benefit
Static EtherChannel	Simple and predictable for a two-link bundle

4. Implementation in Packet Tracer

4.1 Core services

- **Layer 2:** VLANs 10, 20, 30, 40 and 99 exist on CORE-L3SW; VLANs 30 and 99 exist on ACC-READSW. Po1 is a static EtherChannel (`channel-group 1 mode on`) over Gi1/0/23 and Gi1/0/24 on the core and Fa0/23 and Fa0/24 on the access switch, configured as a trunk.
- **Layer 3:** `ip routing` is enabled on CORE-L3SW with an SVI for each VLAN. The core has a default route to RTR-EDGE (10.42.0.1) and the router has a route for 10.42.0.0/16 via the core (10.42.0.2).
- **DHCP:** CORE-L3SW serves ADMIN-POOL, ARCHIVE-POOL and READING-POOL, each with its own default router. The first nine addresses of each subnet are excluded so gateways and future fixed devices keep predictable addresses.
- **Remote management (CR9):** all three devices accept SSH version 2 only (`transport input ssh`, `login local`). RTR-EDGE accepts sessions only from the off-site administrator (access-list 10, host 192.0.2.2). The switches accept sessions only from ACL MGMT-SSH: the off-site administrator, the Administration VLAN and the Management VLAN.

4.2 Fault audit and corrections

Before testing, the implementation was audited against the design. A script checked 52 items of stored configuration and cabling: 36 passed on the original file and 49 on the corrected file, with the remaining 3 being RSA keys generated live during testing. The audit found seven faults, all corrected before evidence was captured.

No.	Fault found	Effect	Correction
1	Reading Room PCs cabled to Fa0/3 and Fa0/4, but only Fa0/1	PCs stayed in VLAN 1 and got no DHCP lease	Fa0/3 and Fa0/4 set as VLAN 30 access ports

No.	Fault found	Effect	Correction
	and Fa0/2 were in VLAN 30		
2	Po1 on ACC-READSW built on Gi0/1 and Gi0/2, but cables are on Fa0/23 and Fa0/24	No working bundle on the access side	Po1 members changed to Fa0/23 and Fa0/24
3	RTR-EDGE Gi0/0/1 had no IP address and neither device had a route to the other	No path between the campus and the edge router	Transit link 10.42.0.0/30 added; core Gi1/0/6 made a routed port; default route on the core and 10.42.0.0/16 route on the router
4	Core and access switch had no enable secret; user tt had privilege 1	SSH login could not reach privileged mode	User tt set to privilege 15 on all three devices
5	Only the router restricted SSH by ACL	Switches accepted SSH from any VLAN, against CR9	ACL MGMT-SSH added to the switch vty lines
6	No saved startup configuration	A reload would erase all configuration	Configuration saved on all three devices
7	RSA keys missing on all devices	SSH cannot start without a key pair	Keys generated on each device

5. Assigned feature: restricted internal web server

Requirement: the HTTP server in VLAN 40 must be reachable only from VLAN 10 (Administration) and VLAN 20 (Archives/Research).

5.1 Implementation

WEB-SRV (10.42.40.10) has its HTTP service enabled. The restriction is enforced on CORE-L3SW by the extended ACL WEB-ACCESS, applied outbound on interface Vlan40. Every inter-VLAN packet towards the servers leaves the core through Vlan40, so one ACL covers every other source, including the Reading Room, the Management VLAN and the off-site administrator.

```
ip access-list extended WEB-ACCESS
 permit tcp 10.42.10.0 0.0.0.255 host 10.42.40.10 eq www
 permit tcp 10.42.20.0 0.0.0.255 host 10.42.40.10 eq www
 permit icmp 10.42.10.0 0.0.0.255 10.42.40.0 0.0.0.255
 permit icmp 10.42.20.0 0.0.0.255 10.42.40.0 0.0.0.255
```

```
 deny ip any 10.42.40.0 0.0.0.255
 permit ip any any
!
interface Vlan40
 ip address 10.42.40.1 255.255.255.0
 ip access-group WEB-ACCESS out
```

5.2 Supporting control for the Reading Room

The Reading Room is public-facing, so ACL READING-ISOLATE is applied inbound on Vlan30 and denies traffic from VLAN 30 to VLANs 10, 20, 40 and 99, while Reading Room PCs still reach their own gateway. This gives two independent layers keeping public terminals away from the server.

5.3 Design decisions

- **Outbound on Vlan40:** one ACL protects the whole server VLAN, and any new user VLAN is blocked from it by default.
- **Ordering:** specific permits first, then an explicit deny for the server subnet, then `permit ip any any`. Without the final permit, the implicit deny would drop every other packet leaving that interface.
- **Host-specific permit:** web access is permitted to 10.42.40.10 only, not the whole VLAN, so a future server is not exposed by accident.
- **ICMP from VLANs 10 and 20:** lets staff test the server with ping. Only port 80 is open for applications; SSH, FTP and other protocols from those VLANs are blocked.
- **Two layers for the Reading Room:** READING-ISOLATE stops Reading Room traffic first; WEB-ACCESS remains the control that protects the server from every other source.

6. Testing and verification

Each test was run in Packet Tracer on the corrected network. Full screenshots for every test are in the Milestone 2 evidence report and the GitHub portfolio.

ID	Test	Result observed	Status
T1	DHCP, Administration	PC-Admin1 10.42.10.10 and PC-Admin2 10.42.10.11, gateway 10.42.10.1	Pass
T2	DHCP, Archives	PC-Archivist1 10.42.20.10 and PC-Archivist2 10.42.20.11, gateway 10.42.20.1	Pass
T3	DHCP, Reading Room	PC-Reading1 10.42.30.11 and PC-Reading2 10.42.30.10, gateway 10.42.30.1	Pass

ID	Test	Result observed	Status
T4	Gateway reachability	All six PCs ping their gateway, 4 of 4, 0% loss	Pass
T5	Inter-VLAN routing	PC-Admin1 to PC-Archivist1: 4 of 4, TTL 127	Pass
T6	EtherChannel and VLANs	Po1(SU) with both member ports (P) on core and access switch; VLANs present	Pass
T7	EtherChannel failover	One cable removed: about 6 pings lost, then recovered without intervention; both ports (P) after reconnecting	Pass
T8	Web access, Administration	PC-Admin1 loads http://10.42.40.10	Pass
T9	Web access, Archives	PC-Archivist1 loads http://10.42.40.10	Pass
T10	Web blocked, Reading Room	PC-Reading1 browser shows Request Timeout	Pass
T11	Ping blocked, Reading Room	0 of 4; Destination host unreachable from 10.42.30.1	Pass
T12	ICMP from Administration	PC-Admin1 ping to server: 4 of 4	Pass
T13	ACL counters	Match counts on WEB-ACCESS and READING-ISOLATE confirm the intended lines are hit	Pass
T14	Reading Room isolation	PC-Reading1 cannot reach server, PC-Admin1 or 10.42.99.1; own gateway still	Pass

ID	Test	Result observed	Status
		replies	
T15	Router to core link	RTR-EDGE and CORE-L3SW ping each other's transit address, 5 of 5	Pass
T16	Routing from off-site PC	Remote-Admin-PC reaches 192.0.2.1 and 10.42.10.1, 4 of 4	Pass
T17	Server shielded from other sources	Remote-Admin-PC and RTR-EDGE cannot reach 10.42.40.10	Pass
T18	SSH from off-site administrator	Reached RTR-EDGE, CORE-L3SW and ACC-READSW at the privileged prompt	Pass
T19	SSH blocked from other sources	Reading Room refused or timed out; Administration allowed to the core by design, refused by the router	Pass
T20	Telnet not usable	Connection opens, then the router closes it at once with no login prompt	Pass
T21	Configuration saved	`write memory` returned [OK] on all three devices	Pass

Honest notes on two results. The expected result for T7 was refined after observing Packet Tracer: failover loses about six pings rather than none. For T20, Packet Tracer closes the Telnet connection immediately instead of printing a refused message. Both are recorded as observed.

Coverage gap. Some ACL lines (for example the ICMP permit for VLAN 20 and the VLAN 30 to VLAN 20 deny) show no match count because no dedicated test exercised them. They mirror the tested VLAN 10 lines and were checked in the configuration.

7. Differences from the Milestone 1 design

The build changed in several places while moving from design to a working network. These are recorded so the report matches the submitted Packet Tracer file.

Area	Milestone 1 design	Final build	Reason
Core switch model	Cisco 3560	Cisco 3650-24PS	Model available and suitable in Packet Tracer 9.0.1
Router to core link	802.1Q trunk, router interface without IP	Routed /30 transit link 10.42.0.0/30 with static routes	Core performs all inter-VLAN routing, so the router needs only one routed path to the campus
Router interface names	Gi0/0 and Gi0/1	Gi0/0/0 and Gi0/0/1	Naming on the ISR4321
WAN side	Internet cloud shown	Remote-Admin-PC connected directly to the router	Simulation simplification
NAT/PAT on RTR-EDGE	Proposed for outbound internet access	Not configured	No internet-side network exists in the simulation; recorded as future work
DHCP pools	Range .10 to .200, Reading Room lease 4 hours	Whole /24 with .1 to .9 excluded; default lease time	Simpler pool definition; lease tuning noted as future work
Reserved addresses	.2 to .9 reserved	.1 to .9 excluded from DHCP	Includes the gateway address in the exclusion
SSH access control	ACL on RTR-EDGE only	MGMT-SSH on both switches as well, plus ACL 10 on the router	Fault 5: switches had accepted SSH from any VLAN, against CR9
Server port	Fa0/1	Gi1/0/1 on the core	Matches the 3650 port layout

8. Security review

- **Segmentation:** public Reading Room traffic is blocked from VLANs 10, 20, 40 and 99 by READING-ISOLATE, and the server is independently protected by WEB-ACCESS.
- **Management plane:** SSH version 2 only, local authentication, and source-restricted vty access on all three devices. Telnet is unusable.
- **Least privilege on the server:** only TCP 80 and ICMP from VLANs 10 and 20; host-specific permit.
- **Known weaknesses in this simulation:** the enable secret is not set and the console line has no password, which is acceptable inside a simulation but would be set in production. Passwords are

not recorded in this report or in the repository. The MGMT-SSH list deliberately allows the Administration VLAN to manage the switches.

9. Limitations and future work

- **Failover is short, not zero:** removing one EtherChannel cable cost about six pings while the bundle re-formed.
- **Single points of failure:** one edge router and one core switch remain. A second core switch with stacking or a first-hop redundancy protocol would remove this.
- **Static choices:** static routing and a static EtherChannel were chosen for simplicity. LACP would negotiate the bundle and detect a misconfigured partner, and a dynamic routing protocol would be considered if more routers were added.
- **Internet access:** NAT/PAT, an ISP link and public Reading Room internet access are not modelled.
- **DHCP tuning:** shorter leases for the transient Reading Room population.
- **Further hardening:** enable secrets, console passwords, port security on access ports, and logging.

10. Reflection

The most valuable lesson was the audit step. Building the network from the design was not enough: a configuration check against the design exposed seven faults, including cabling and port mismatches that would have been hard to spot from symptoms alone. Testing the failure case (T7) showed that redundancy reduces disruption rather than eliminating it, and recording that honestly was more useful than forcing the test to match the plan. Placing the web-server ACL outbound on Vlan40 turned out to be a cleaner design than filtering at every user VLAN, because one list protects the server from every source, including ones added later.

11. Submission contents

Item	Purpose
Packet Tracer project (.pkt)	Working implementation with RSA keys and local users in place
GitHub portfolio	Configurations, screenshots, planning, testing and change history
This technical report	Requirements, design, implementation, assigned feature, testing and reflection
Inset video demonstration (15 to 20 minutes)	Walk-through of the working network and design decisions

Appendix A: Marking criteria mapping

Rubric component (weight)	Where it is evidenced

Rubric component (weight)	Where it is evidenced
Client requirements and network design (15%)	Sections 2, 3 and 7
Packet Tracer implementation and functionality (35%)	Sections 4 and 6; the .pkt file
Assigned feature (20%)	Section 5, tests T8 to T14 and T17
GitHub portfolio (15%)	Repository link on the title page; Section 11
Video demonstration and technical defence (15%)	Section 11; the video